

Study Notes: DHCPv4, NAT, and PAT

1 Dynamic Host Configuration Protocol v4 (DHCPv4)

1.1 Overview & Allocation Modes

DHCPv4 automates IP address assignment on IPv4 networks.

- **Static Allocation:** Used for infrastructure devices requiring fixed, unchangeable IP addresses (e.g., Web Servers, Printers).
- **Dynamic Allocation:** Automatically fetches and leases IP addresses to end-user devices for a limited duration.

Implementation Scope: DHCP can run as a dedicated enterprise server or as a lightweight service integrated into home routers.

1.2 The DORA Process

DHCP operations follow a four-step sequence known as **DORA**:

1. Discover (Client → Server | Broadcast)

A newly connected client broadcasts to locate available DHCP servers.

- **Source MAC:** Client MAC | **Destination MAC:** FF:FF:FF:FF:FF:FF
- **Source IP:** 0.0.0.0 | **Destination IP:** 255.255.255.255
- **Source Port:** 68 (Client) | **Destination Port:** 67 (Server)

2. Offer (Server → Client | Unicast)

The server reserves an IP and unicasts configuration settings to the client.

- **Offered Parameters:** Assigned IP, Subnet Mask, Default Gateway, DNS Server, Lease Time.
- **Addressing:** Destination MAC matches the client's MAC. Destination IP transitions target from 0.0.0.0 to the newly offered IP.

3. Request (Client → Server | Broadcast)

The client broadcasts its intent to accept the offered IP address.

- **Purpose of Broadcast:** Informs all other DHCP servers on the network that their offers were declined so they can release their reserved IPs back to their pools.

4. Acknowledgment / ACK (Server → Client | Unicast)

The selected server acknowledges the request and finalizes the lease binding.

1.2.1 Example: DHCP Offer Packet Header

Header Field	Value
Source MAC	00:1A:2B:3C:4D:5E (DHCP Server MAC)
Destination MAC	AA:BB:CC:11:22:33 (Client MAC)
Source IP	192.168.1.1 (DHCP Server IP)
Destination IP	192.168.1.50 (Offered IP)
Payload Configuration	IP: 192.168.1.50, Subnet: 255.255.255.0, Gateway: 192.168.1.1

1.3 Lease Renewal Timeline

- **50% Duration (T_1):** Client sends a **Unicast** DHCP Request directly to the original server to renew the lease.
- **77.5% Duration (T_2):** If no ACK is received by 77.5% of the total lease duration, the client attempts to rebind by resending requests.
- **Expiration:** If no ACK is received before the lease time reaches 100%, the IP address is surrendered immediately.

1.4 DHCP Relay

- **Problem:** Routers drop broadcast packets (255.255.255.255) by default. If the DHCP server resides on a different network segment, client Discover packets fail.
- **Solution:** Configure the router interface (Default Gateway) as a **DHCP Relay Agent**. The router converts broadcast DHCP requests into unicast packets addressed directly to the remote DHCP server.
- **Client Resolution:** Address Resolution Protocol (ARP) is used to map return packets back to the original client's hardware address.

2 Network Address Translation (NAT)

2.1 Overview & Operation

NAT conserves IPv4 address space by allowing multiple private network devices to share public IPv4 addresses.

1. An outbound packet leaving a private IP address reaches the edge router.
2. The router translates the internal **Private IP** to an available **Public IP**.
3. The router logs this mapping in its internal **NAT Table**.
4. Inbound response packets reaching the router's public IP are mapped back to the corresponding private IP using the NAT Table.

Resource Constraint: If no public IP addresses are available in a dynamic pool, host traffic must wait until an IP is freed.

2.2 Disadvantages of NAT

1. **Loss of Traceability:** End-to-end IP traceability is obscured due to continuous header translation.
2. **Latency:** Header modification introduces processing delay at the router level.

3 Port Address Translation (PAT / NAT Overload)

3.1 Mechanism

PAT allows thousands of internal hosts to share a **single public IP address** simultaneously by appending unique source port numbers to distinguish sessions.

3.2 Special Handling Scenarios

- **Port Collisions:** If two internal devices initiate connections using the same source port number, PAT assigns the next available public port to the second host.
- **Protocols Without Ports (e.g., ICMP):** PAT uses a **Query ID** in place of a port number to track translation table entries.
- **Inbound Connections to Private IPs (Port Forwarding):** To allow external access to an internal server (e.g., Web Server or Load Balancer), static rules are configured on the router mapping an incoming `Public_IP:Port` combination to a specific internal `Private_IP:Port`.

3.2.1 Example: PAT Translation Table State

Inside Local (Private)	Inside Global (Public)	Outside Global (Destination)	Protocol
192.168.1.10:5000	203.0.113.1:10001	140.82.112.4:443	TCP
192.168.1.11:5000	203.0.113.1:10002	140.82.112.4:443	TCP

4 Student Review & Practice Questions

1. **DHCP Ports:** Why does DHCP operate over UDP ports 67 (server) and 68 (client) rather than TCP?
2. **DORA Logic:** Why must the DHCP Request packet (Step 3) be sent as a broadcast if the client already received a unicast Offer from a specific server?
3. **Lease Renewal Math:** A device receives a DHCP lease with a duration of 8 hours (28,800 seconds). At what exact timestamps (in hours and minutes) will T_1 renewal and T_2 rebind actions occur?
4. **NAT vs PAT:** Explain how PAT handles a scenario where host A (10.0.0.2:8080) and host B (10.0.0.3:8080) both attempt to access the exact same external server at the same time using a single public IP.
5. **DHCP Relay:** What specific issue occurs when a DHCP client boots on Subnet A while the DHCP server is on Subnet B, and how does the Default Gateway resolve this issue?